

Time-series Foundation Model Evaluation

Executive Summary

Thesis project

Evidence checked through 26 August 2026

Decision snapshot

Chronos-2 consistently improves on the non-trainable baselines for the three completed hourly datasets. This evidence supports Chronos-2 for the evaluated univariate hourly settings, but does not yet support a conclusion for daily Exchange Rate, TIME datasets, covariates, or the four-model foundation comparison.

1 Evidence inventory

Result object	Count
Current hourly datasets	3
Current dataset-setting configurations	9
Current model configurations	45
Seeds per configuration	1
Completed TIME evaluations	0

2 Obtained evidence

The completed full-univariate snapshot evaluated Electricity, Solar, and Traffic at 168:24, 336:48, and 504:168 with seed 1 and stride 512. Chronos-2 beat the best persistence, expected, repeat, or weekly-lookback baseline in nMSE on all nine configurations. Its mean nMSE improvement over the best baseline was 31.335% on Electricity, 43.099% on Solar, and 37.394% on Traffic.

These are single-seed, sparsely sampled results. The warning about an unset deterministic CUDA workspace variable further limits a claim of exact GPU reproducibility. The former Exchange Rate results used hourly-scale $L-H$ values on daily data and are excluded from the current protocol.

3 Next evidence

Resubmit full univariate under the cadence-aware profile. Exact completed hourly identities remain reusable; the run must add daily Exchange Rate at 7:1, 14:2, and 30:7, evaluate every feasible prepared TIME dataset, and regenerate the mixed-cadence report. Then complete full controls, covariates, and foundation-model families before making claims about those questions.